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Assignment 3

1. $x[n] = \{1, 2, 1\}$, $n = 0, 1, 2$ $\quad h[n] = \{1, -1, 2\}$, $n = 0, 1, 2$

a). $y[n] = x[n] * h[n]$

$$= \sum_{m=-\infty}^{\infty} x[m]\, h[n-m]$$

$$y[0] = \sum_{m=-\infty}^{\infty} x[m]\, h[-m]$$

$$= x[0]\,h[0] + x[1]\,h[-1] + x[2]\,h[-2] + \cdots$$

$$= \phi + 0 + 0 + \cdots$$

$$= 1$$

$$y[1] = \sum_{m=-\infty}^{\infty} x[m]\, h[1-m]$$

$$= x[0]\,h[1] + x[1]\,h[0] + x[2]\,h[-1] + 0$$

$$= 1 \times -1 + 2 \times 1 + 0$$

$$= 1$$

$$y[2] = \sum_{m=-\infty}^{\infty} x[m]\, h[2-m]$$

$$= x[0]\,h[2] + x[1].h[1] + x[2]\,h[0] + 0$$

$$= 1 \times 2 + 2 \times -1 + 1 \times 1$$

$$= 1$$

$$y[3] = \sum_{m=-\infty}^{\infty} x[m]\, h[3-m]$$

$$= x[0]\,h[3] + x[1]\,h[2] + x[2].h[1] + x[3]\,h[0] + 0$$

$$= 1 \times 0 + 2 \times 2 + 1 \times -1 + 0$$

$$= 3$$

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$$y[4] = \sum_{m=-\infty}^{\infty} x(m)\, h[4-m]$$

$$= x[0].h(4) + x(1).h(3) + x(2).h(2) + x(3).h(1) + 0$$

$$= 0 + 0 + 1\times 2$$

$$= 2$$

$$y[n] = \{1, 1, 1, 3, 2\} \quad n = 0, 1, 2, 3, 4$$

b)

n	$\sum_{m=-\infty}^{\infty} x(m)\, h(n-m)$	$y(n)$
0	1×1	1
1	$1\times -1 + 2\times 1$	1
2	$1\times 2 + 2\times -1 + 1\times 1$	1
3	$1\times 0 + 2\times 2 + 1\times -1$	3
4	1×2	2

2).

a). $h_1(n) = \{1, 1\}, \; n = 0, 1 \qquad h_2(n) = \{1, -1, 1\} \quad n = 0, 1, 2$

$$h_{eq}(n) = h_1(n) * h_2(n)$$

$$= \sum_{k=-\infty}^{\infty} h(k)\, h(n-k)$$

$$h(0) = h_1(0)\, h_2(0)$$

$$= 1$$

$$h_{eq}(n) = \{1, 1, 0, 0\} \quad n = 0, 1, 2, 3$$

$$h(1) = h_1(0)\, h_2(1) + h_1(1).h_2(0)$$

$$= -1 + 1$$

$$= 0$$

$$h(2) = h_1(0)\, h_2(2) + h_1(1)\, h_2(1) + 0$$

$$= 1 + -1$$

$$= 0$$

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duration of $h_{eq}[n] = n_1 + n_2 - 1$

$= 2 + 3 - 1$

$= 4$